

5 (a) (i) sum of currents into a junction = sum of currents out of junction	B1 [1]
(ii) charge	B1 [1]
(b) (i) $\Sigma E = \Sigma IR$ $20 - 12 = 2.0(0.6 + R)$ (not used 3 resistors 0/2) $R = 3.4 \Omega$	C1 A1 [2]
(ii) $P = EI$ $= 20 \times 2$ $= 40 \text{ W}$	C1 A1 [2]
(iii) $P = I^2 R$ $P = (2)^2 \times (0.1 + 0.5 + 3.4)$ $= 16 \text{ W}$	C1 A1 [2]
(iv) efficiency = useful power / output power $24 / 40 = 0.6$ or $12 \times 2 / 20 \times 2$ or 60%	C1 A1 [2]

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6 (a) (i) diffraction bending/spreading of light at edge/slit

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